ATHLETICS PERFORMANCE ANALYTICS

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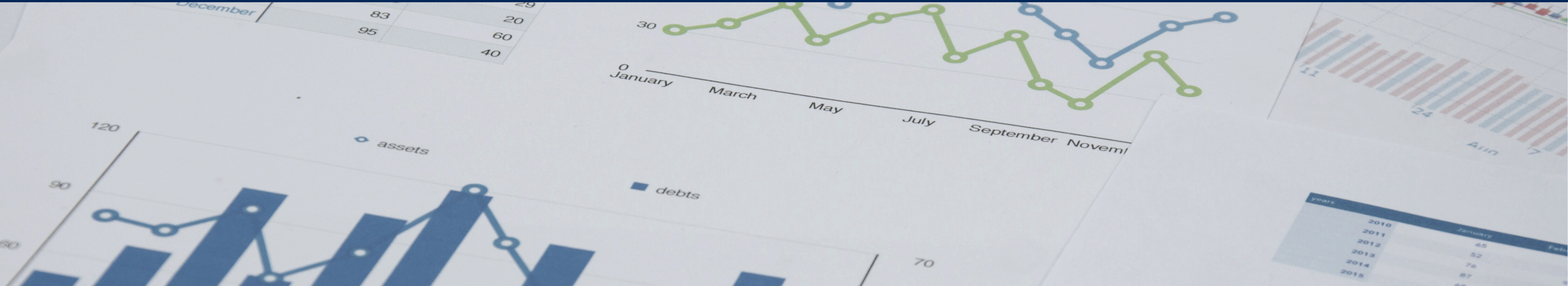


RESEARCH QUESTION & PURPOSE

Research Question:

How do force-plate and GPS metrics change over time, and can they identify fatigue or abnormal workload?

Purpose:

Analyze metrics over time to support athlete readiness and training decisions.

- Inconsistent monitoring of mechanical load
- Highlights the need for standardized multi-system data collection
- Montoring fatigue, workload, & performance readiness

WHY THIS MATTERS

Selected Metrics & Why

Hawkins Force Plate:

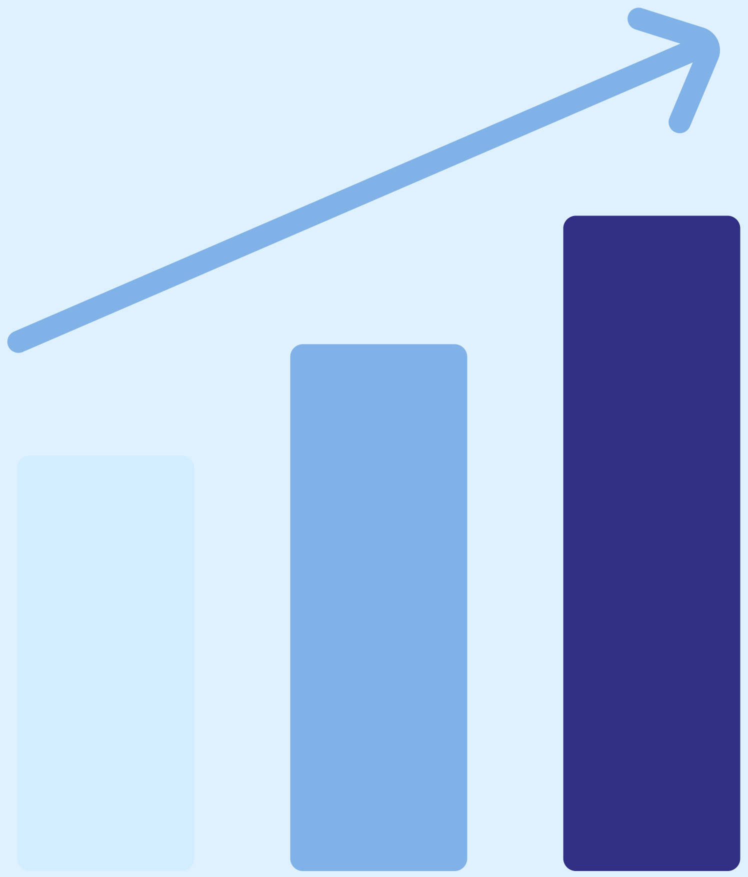
- Jump Height
- RSI
- Peak Propulsive Force

Kinexon GPS:

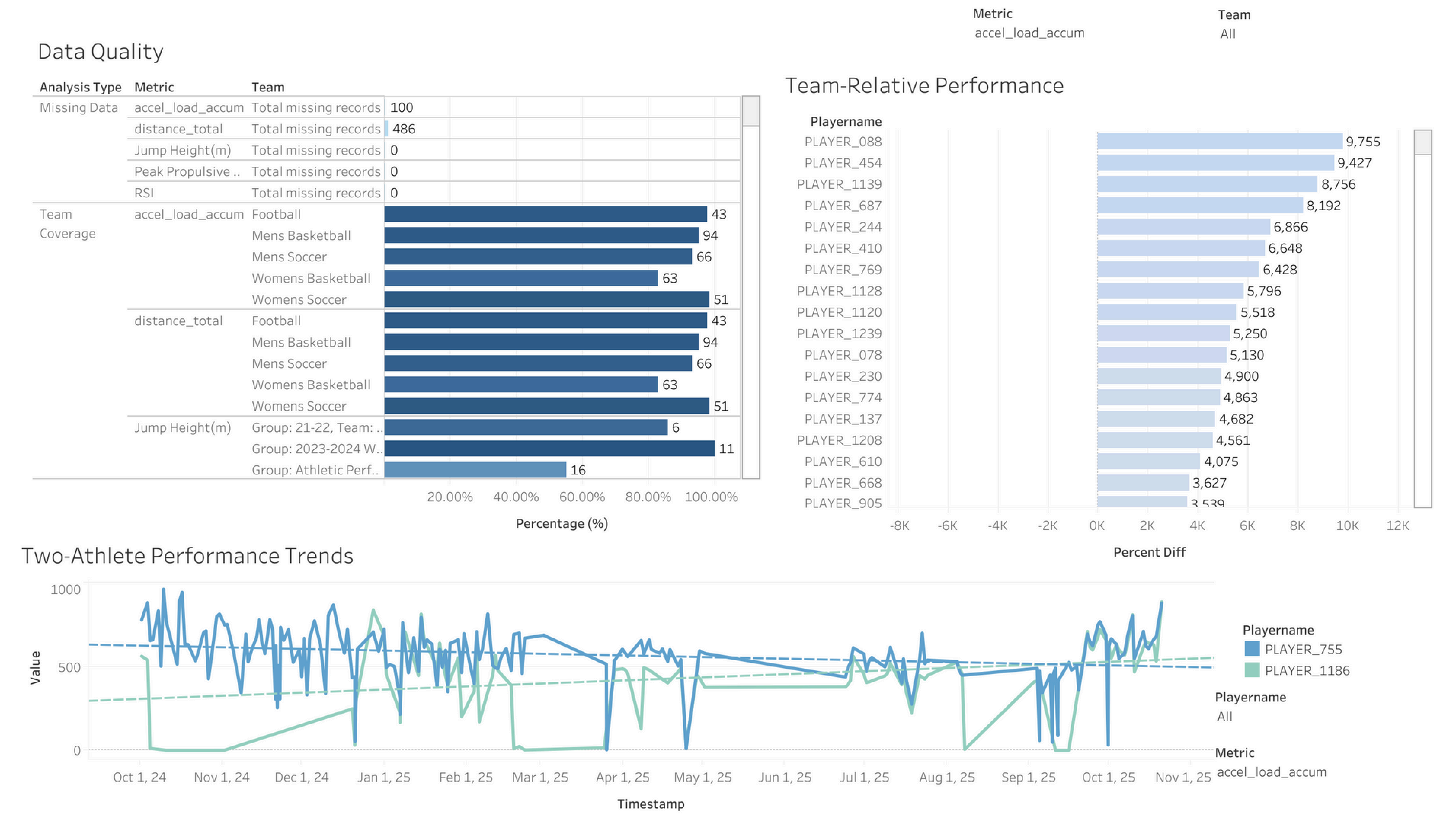
- Accumulated Acceleration Load
- Total Distance

Why These Metrics?

- Capture neuromuscular readiness + mechanical load



DASHBOARD



Metric
accel_load_accum
Team
All
Data Quality
Analysis Type
Metric
Team
Missing Data
accel_load_accum
Total missing records
100
distance_total
Total missing records
486
Jump Height(m)
Total missing records
0
Peak Propulsive ..
Total missing records
0
RSI
Total missing records
0
Team Coverage
accel_load_accum
Football
43
Mens Basketball
94
Mens Soccer
66
Womens Basketball
63
Womens Soccer
51
distance_total
Football
43
Mens Basketball
94
Mens Soccer
66
Womens Basketball
63
Womens Soccer
51
Jump Height(m)
Group: 21-22, Team: ..
6
Group: 2023-2024 W..
11
Group: Athletic Perf..
16
20.00%
40.00%
60.00%
80.00%
100.00%
Percentage (%)
Team-Relative Performance
Playername
PLAYER_088
9,755
PLAYER_454
9,427
PLAYER_1139
8,756
PLAYER_687
8,192
PLAYER_244
6,866
PLAYER_410
6,648
PLAYER_769
6,428
PLAYER_1128
5,796
PLAYER_1120
5,518
PLAYER_1239
5,250
PLAYER_078
5,130
PLAYER_230
4,900
PLAYER_774
4,863
PLAYER_137
4,682
PLAYER_1208
4,561
PLAYER_610
4,075
PLAYER_668
3,627
PLAYER_905
3,539
-8K
-6K
-4K
-2K
0K
2K
4K
6K
8K
10K
12K
Percent Diff
Two-Athlete Performance Trends
1000
500
0
Value
Oct 1, 24
Nov 1, 24
Dec 1, 24
Jan 1, 25
Feb 1, 25
Mar 1, 25
Apr 1, 25
May 1, 25
Jun 1, 25
Jul 1, 25
Aug 1, 25
Sep 1, 25
Oct 1, 25
Nov 1, 25
Timestamp
Playername
PLAYER_755
PLAYER_1186
Playername
All
Metric
accel_load_accum

KEY FINDINGS

WORKLOAD METRICS: KINEXON

Accumulated Acceleration Load

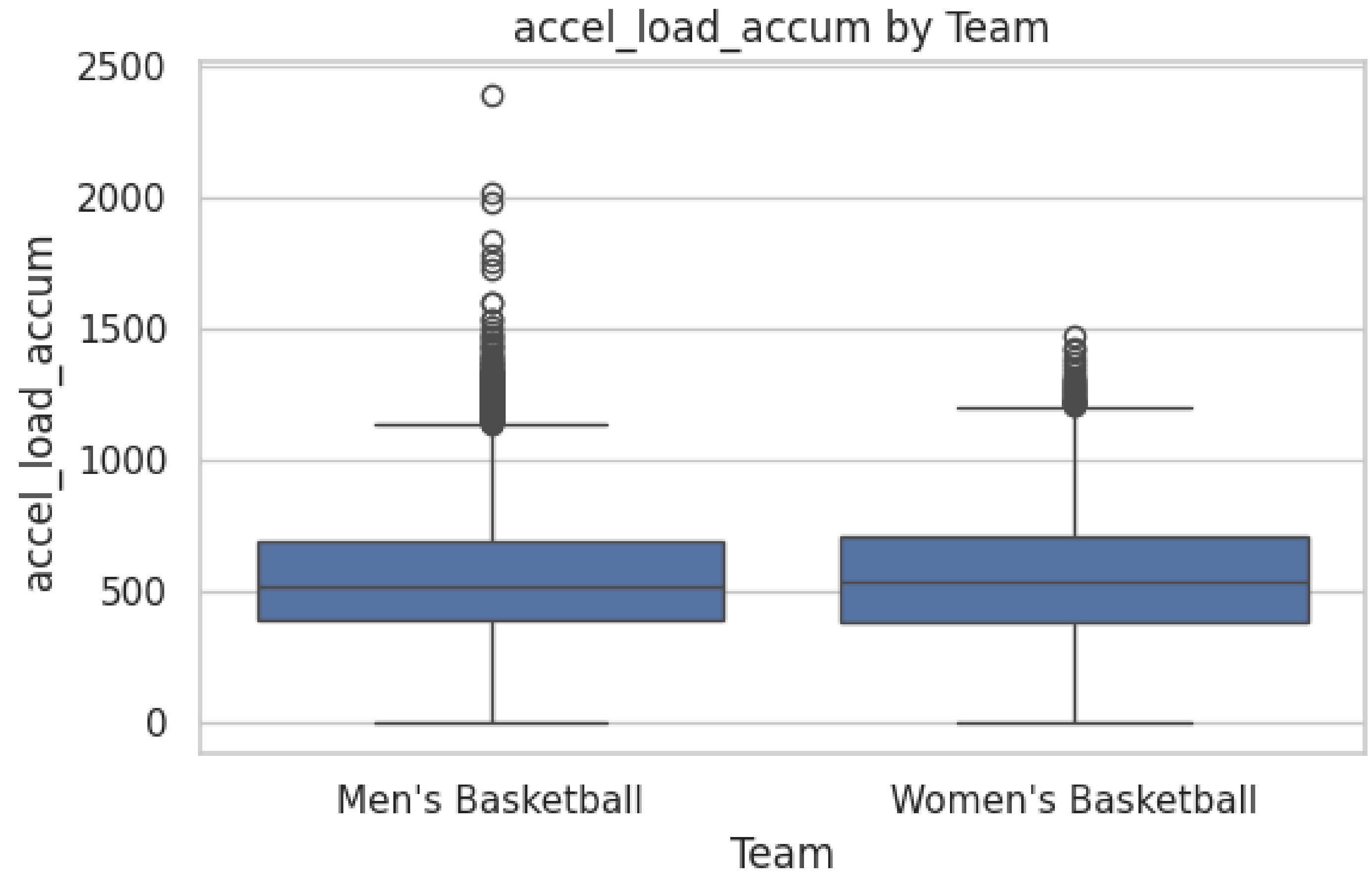
- Men: 9,224 samples
- Women: 8,395 samples
- $t = -2.84$, $p = .0045$ → Women slightly higher

Total Distance

- Men: 9,224
- Women: 8,395
- $t = 1.63$, $p = .104$ → No significant difference

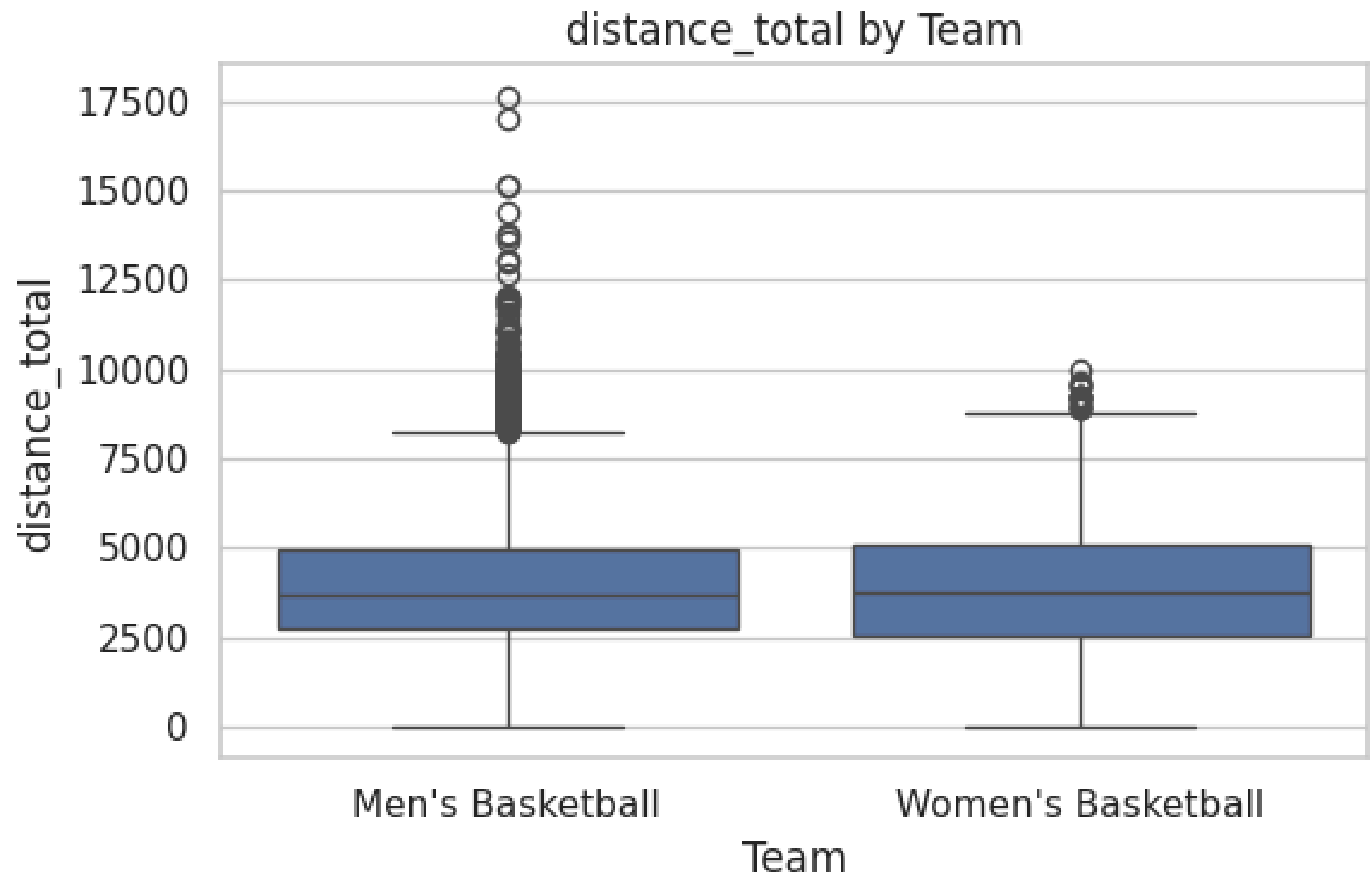
Generated Box Plots

```
Metric: accel_load_accum
Rows for this metric: 17619
```




```
Samples for Men's Basketball: 9224
Samples for Women's Basketball: 8395
t-statistic: -2.83537950764231
p-value   : 0.004582353844008908
```

```
Metric: distance_total
Rows for this metric: 17619
```




```
Samples for Men's Basketball: 9224
Samples for Women's Basketball: 8395
t-statistic: 1.6259024101388189
p-value   : 0.10398830414748231
```

JUMP PERFORMANCE METRICS: HAWKINS

Jump Height (m)

Men: 920 samples

- Women: 613 samples
- $t = 34.77$, $p < 1e\text{-}190$ → Men higher

Peak Propulsive Force (N)

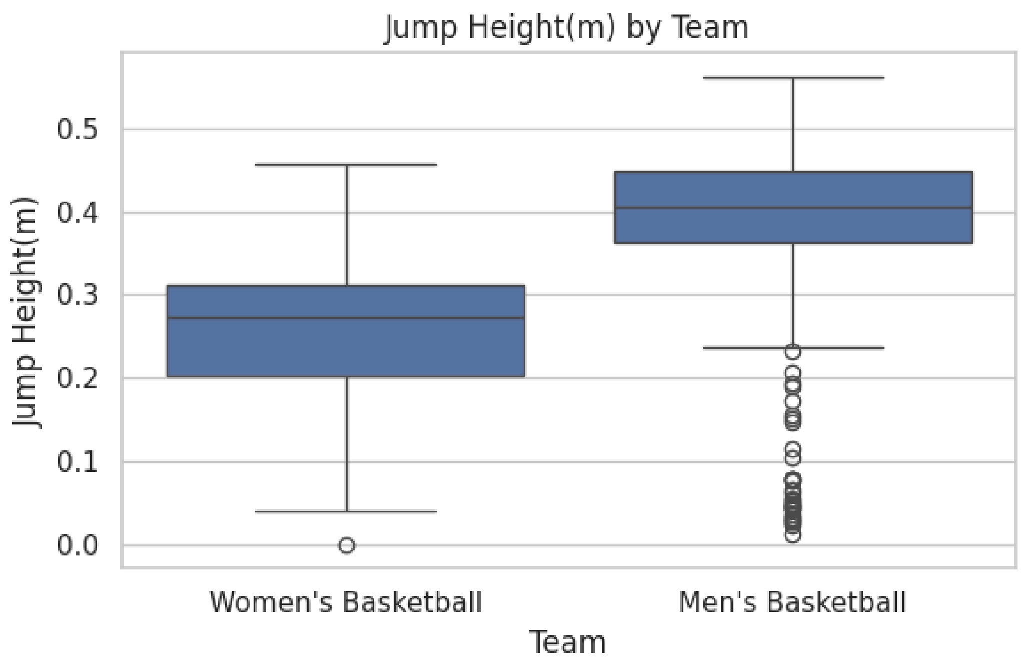
- Men: 920
- Women: 613
- $t = 35.39$, $p < 1e\text{-}193$ → Men higher

RSI

- Men: 917
- Women: 613
- $t = 18.12$, $p < 1e\text{-}65$ → Men higher

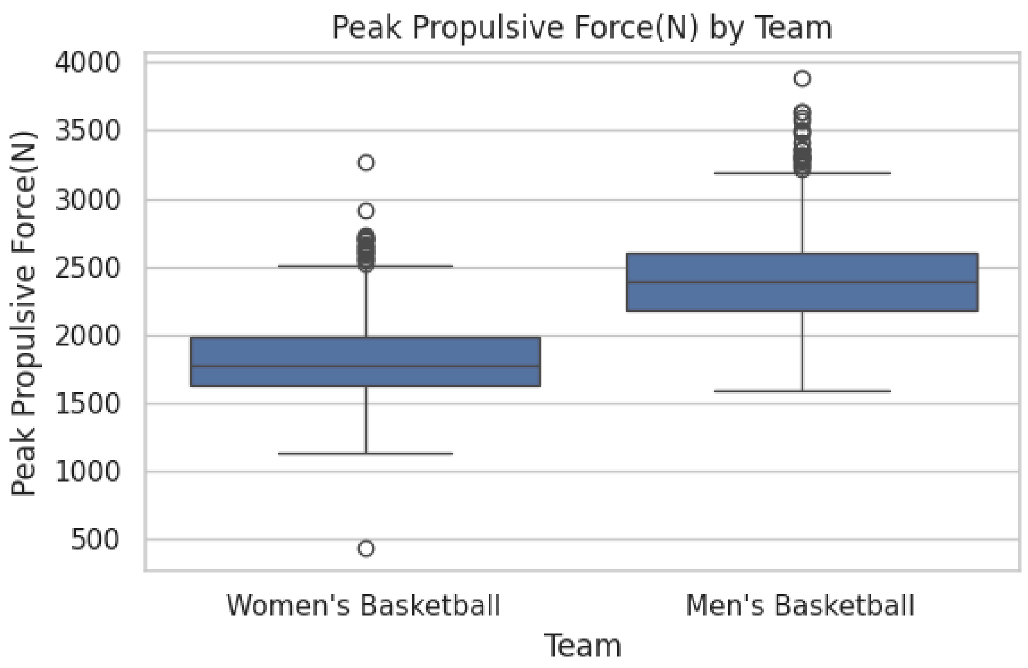
Generated Box Plots

```
Metric: Jump Height(m)
Rows for this metric: 1533
```



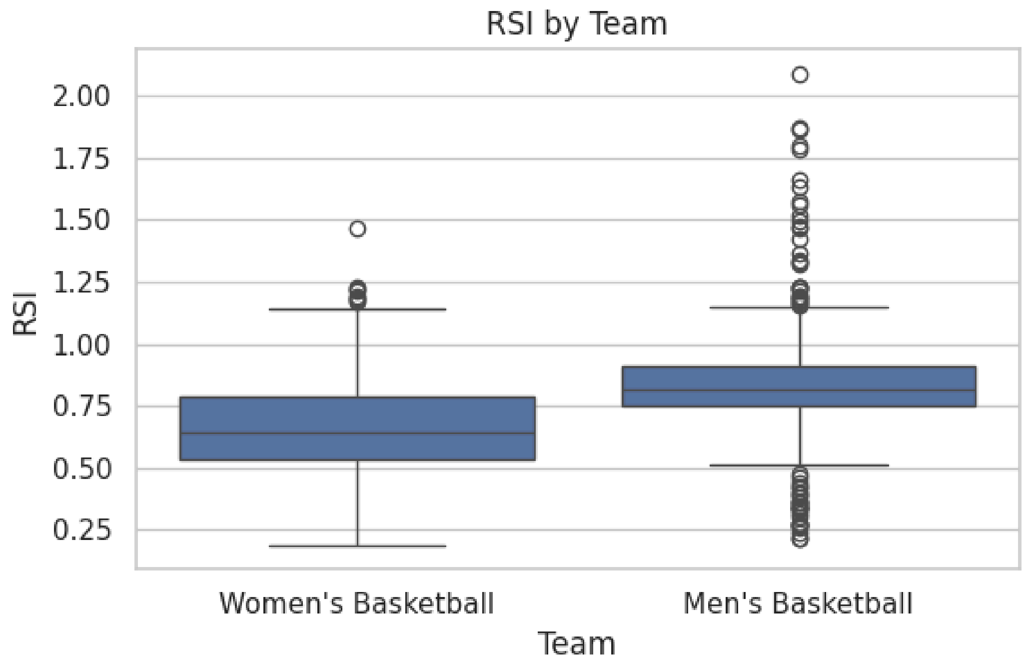

```
Samples for Men's Basketball: 920
Samples for Women's Basketball: 613
t-statistic: 34.772293380154245
p-value    : 1.6447483596524628e-190
```

```
Metric: Peak Propulsive Force(N)
Rows for this metric: 1533
```




```
Samples for Men's Basketball: 920
Samples for Women's Basketball: 613
t-statistic: 35.38584367728158
p-value    : 6.375321507760857e-194
```

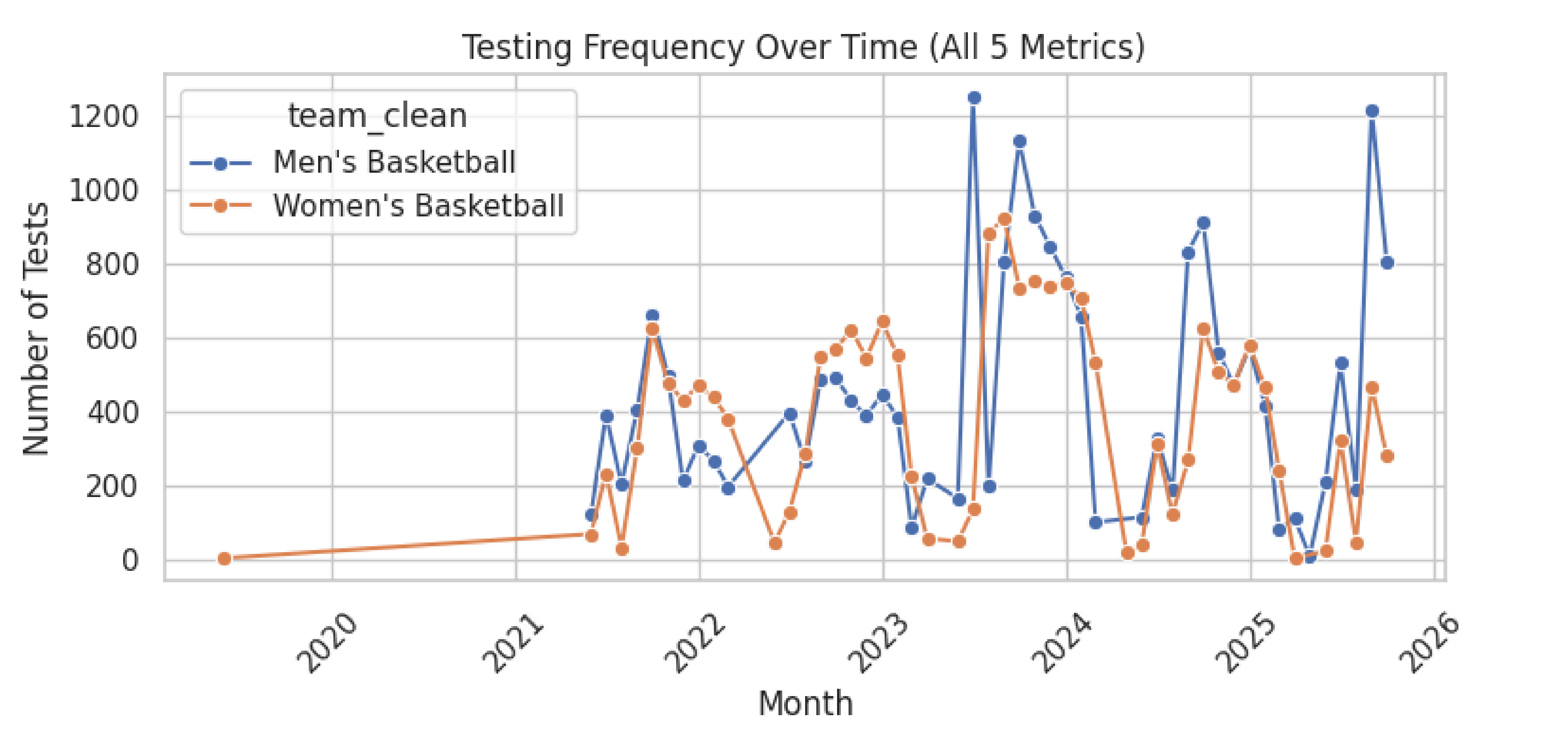
```
Metric: RSI
Rows for this metric: 1530
```



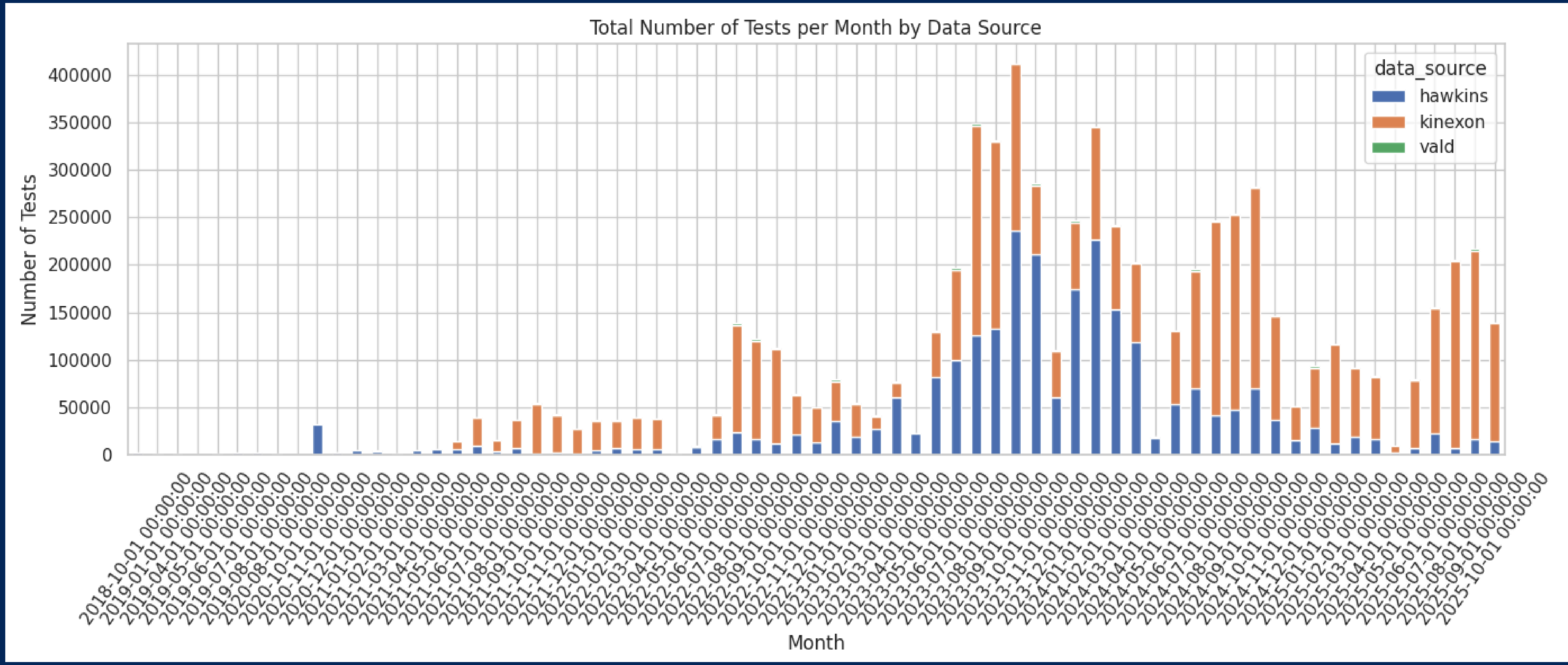

```
Samples for Men's Basketball: 917
Samples for Women's Basketball: 613
t-statistic: 18.123894622219066
p-value    : 1.5382541519022726e-65
```

TESTING FREQUENCY OVER TIME

- Testing rises during the competitive season and drops during summer and winter breaks.
- Higher peaks for Women's Basketball
    - Higher WBB testing volume November 2021



OVERALL TRENDS ACROSS ALL DATA SOURCES



- Kinexon becomes dominant after 2022
- Hawkins steady but lower volume
- VALD minimal
- Sharp increase in overall monitoring 2023–2024

JUMP METRICS:

Our Results vs. Published Norms

Men's team had much higher jump height, peak propulsive force, and RSI than women's

Differences were statistically huge (very small p-values)

Our jump heights, propulsive forces, and RSI values all fall inside published NCAA/elite ranges

This matches prior work showing men have greater lower-body power, muscle mass, and tendon stiffness and typically jump higher

WORKLOAD:

Accel Load & Distance vs Evidence

- **Total distance per session was not significantly different between men and women**
- **This fits prior work showing distance is a volume metric and often looks similar across roles/teams**

- Accumulated acceleration load was slightly higher in women (statistically significant, but small effect)
- This supports literature that accel-based metrics capture “hidden” high-intensity work (stop–go, changes of direction, impacts) that distance alone misses

HOW THE DATA FITS AND EXTENDS

With the Literature

Monitoring Patterns & Where We Add Value:

- Testing frequency follows NCAA seasons (more in pre-/in-season, dips in breaks)
- Use of Hawkins + Kinexon + VALD matches trends toward integrated monitoring
- Our data helps fill literature gaps:
    - Limited work on women's teams
    - Few studies combining jump metrics (CMJ, RSI) with load metrics (accel load, distance)
- Our analysis directly compares men vs. women and links performance + workload in one dataset

PRACTICAL APPLICATIONS

Performance Flagging System

- **CMJ Height / RSI:** Flag when values drop **≥10% below rolling baseline**.
- **AAL (Accumulated Acceleration Load):** Flag when a session exceeds **150% of the athlete's rolling weekly average**.
- **Total Distance:** Flag when daily/session distance is **<80% of team and position-specific norms.**

What This System Provides

- Detects neuromuscular fatigue, acute mechanical load spikes, and abnormal workload reductions.
- Gives practitioners a simple, individualized, and research-supported tool for monitoring readiness and recovery.

RECOMMENDATIONS FOR COACHES/TRAINERS

- Use ≥10% CMJ/RSI drops as a cue to lower intensity or emphasize recovery strategies.
- Treat AAL spikes (>150%) as red-flag sessions that may require modified loading or additional recovery time.
- Investigate <80% total distance days for signs of fatigue, limited engagement, or emerging injury.
- Integrate threshold checks into weekly planning to support consistent, objective decision-making.

ADDRESSING THE GAP

- Prior research often examines metrics individually, but readiness is multidimensional.
- Flagging system integrates neuromuscular, mechanical, and volume-based data into one practical framework.
- Provides a comprehensive, real-world monitoring model that helps staff identify performance issues earlier and more accurately than relying on any single measure.

LIMITATIONS & FUTURE WORK

Data Challenges

- Team names in the database were inconsistent and required cleaning before analysis.
- some metrics showed uneven availbility across teams due to differences in equipmentment usage.
- Different data sources (Hawkins, Kinexon, Vald) had highly unequal test volumes.

Future Work

- improve consistency across teams and data sources
- increase testing frequency to strengthen trend detection

Q&A

As we continue refining our flagging system, do you recommend any specific statistical or coding approaches to validate whether our thresholds are too sensitive or not sensitive enough?